# THE CARTESIAN COORDINATE SYSTEM

## INTRODUCTION TO ANALYTIC GEOMETRY

*prepared by:*

**Gyro A. Madrona**

Electronics Engineer

# GRADING SYSTEM

# FINAL GRADE

30%

PRELIM

30%

MIDTERM

**40%**

ENDTERM

# TERM GRADE

40%

SEATWORK

60%

EXAM

# PENALTY

0

ABSENT

3

45-min LATE

5 max

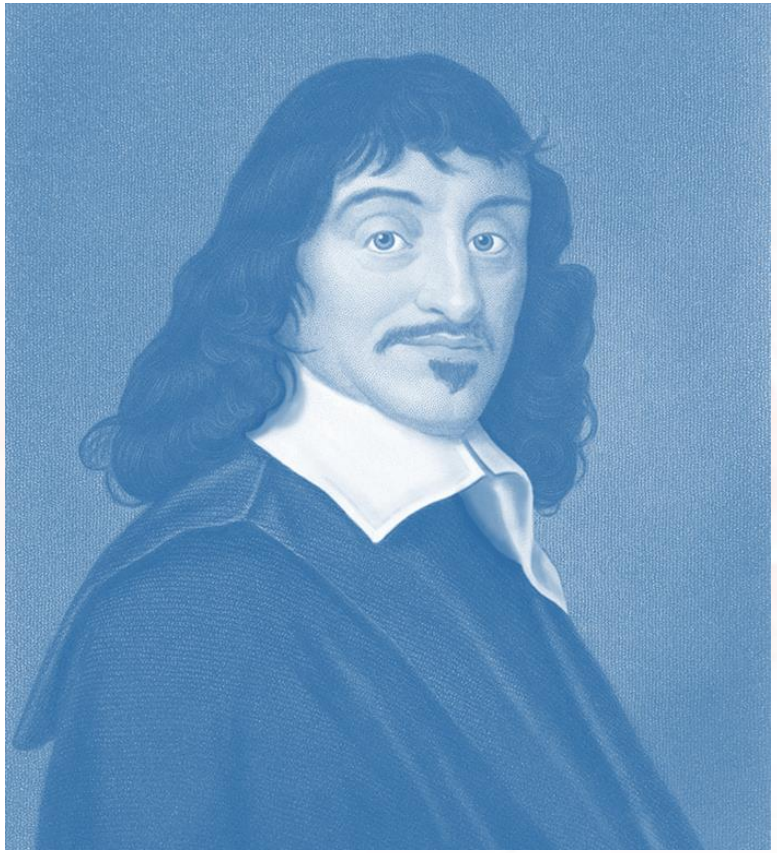
15-min LATE

## TOPIC OUTLINE

### The Cartesian Coordinate System

- Rectangular Coordinates
- Polar Coordinates

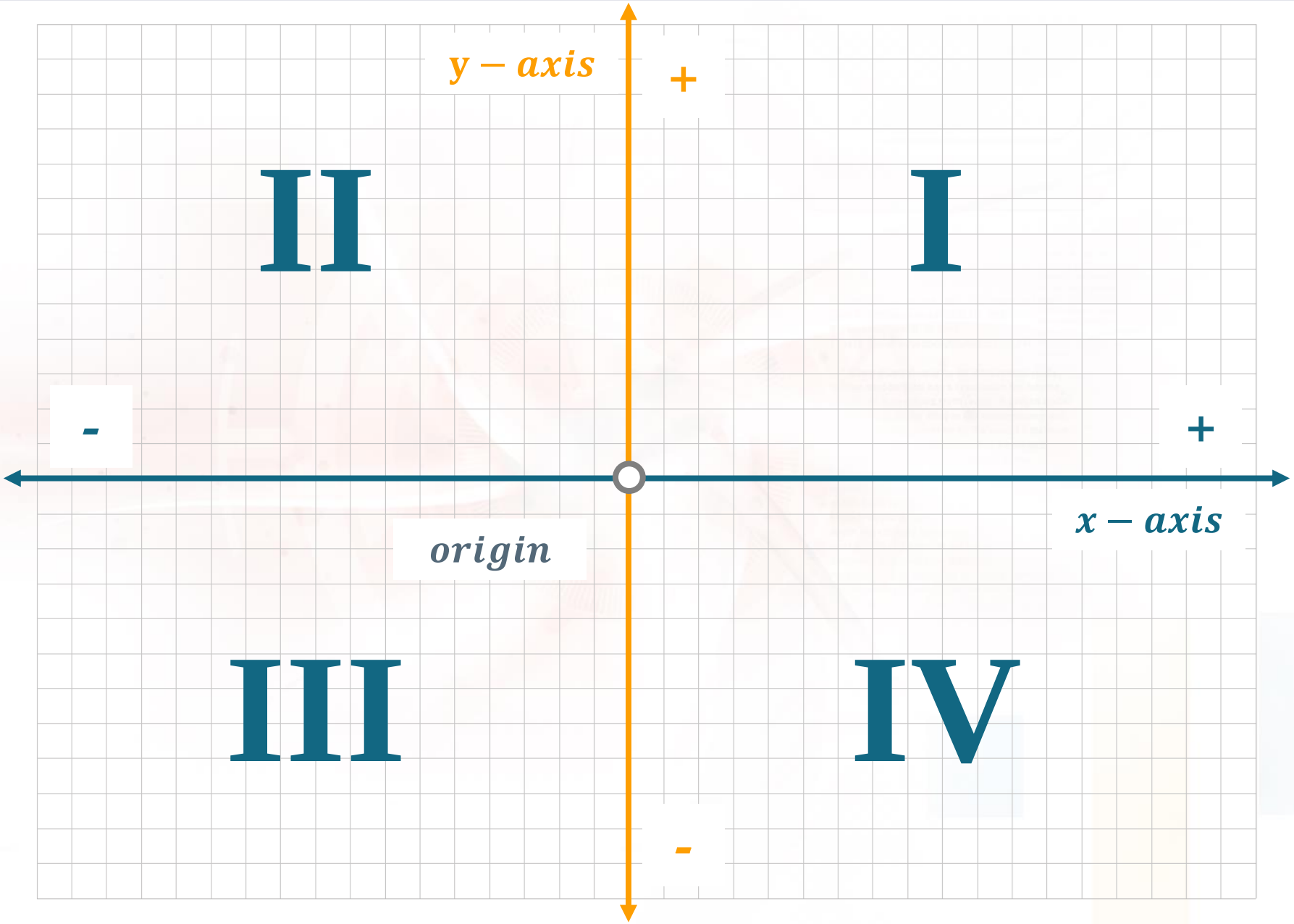
# THE CARTESIAN COORDINATE SYSTEM



René Descartes

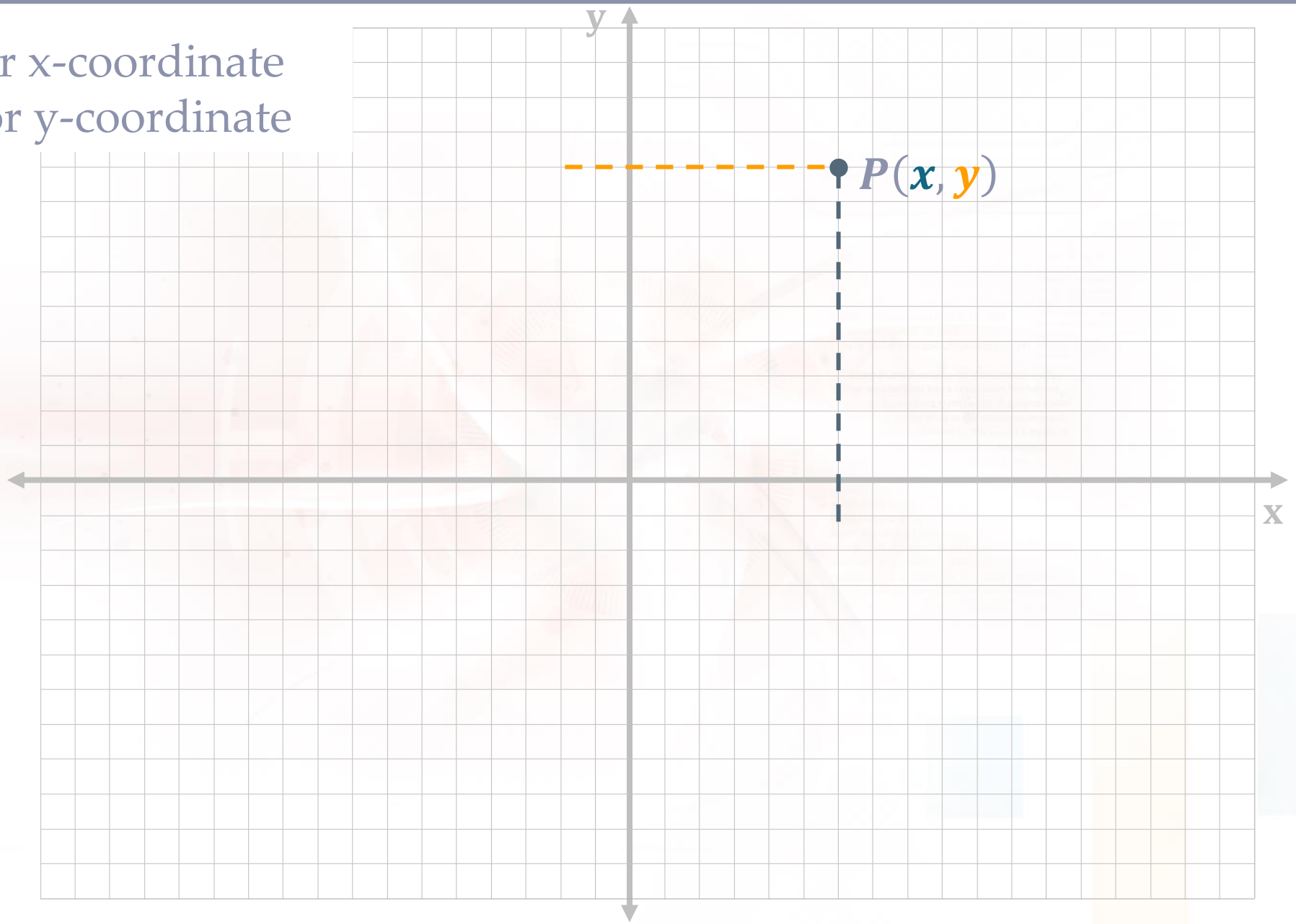
The story is that Descartes was lying in bed late one morning and saw a fly on the ceiling above him. As he watched the fly, he thought about how the **fly's location** could be described using the **edges** of the room for **reference**.

# CARTESIAN PLANE

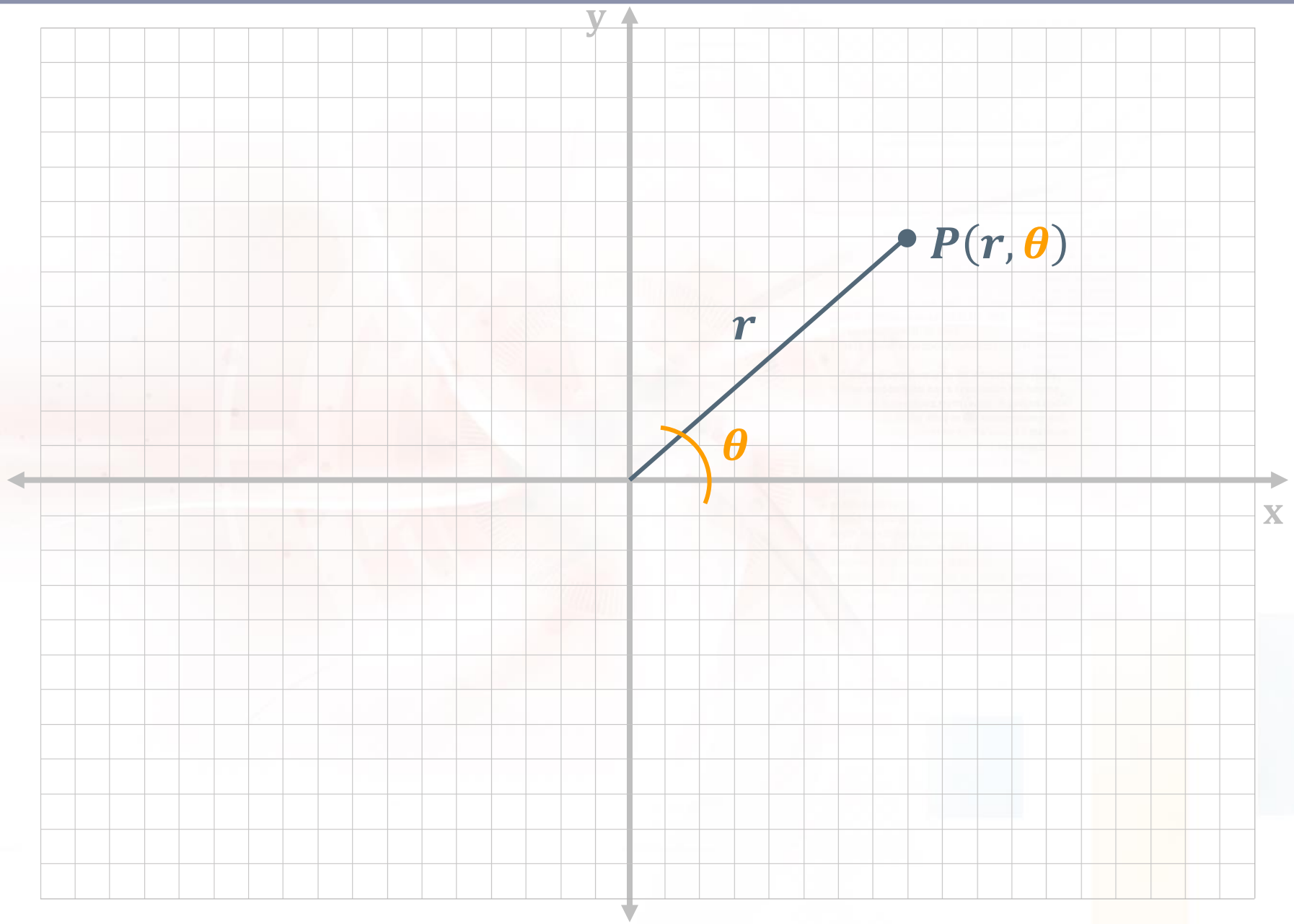


# RECTANGULAR COORDINATES

**abscissa** or x-coordinate
**ordinate** or y-coordinate



# POLAR COORDINATES



Determine the **rectangular** and **polar** coordinates of the given point

$$r_A^2 = x^2 + y^2$$

$$r_A^2 = 8^2 + 7^2$$

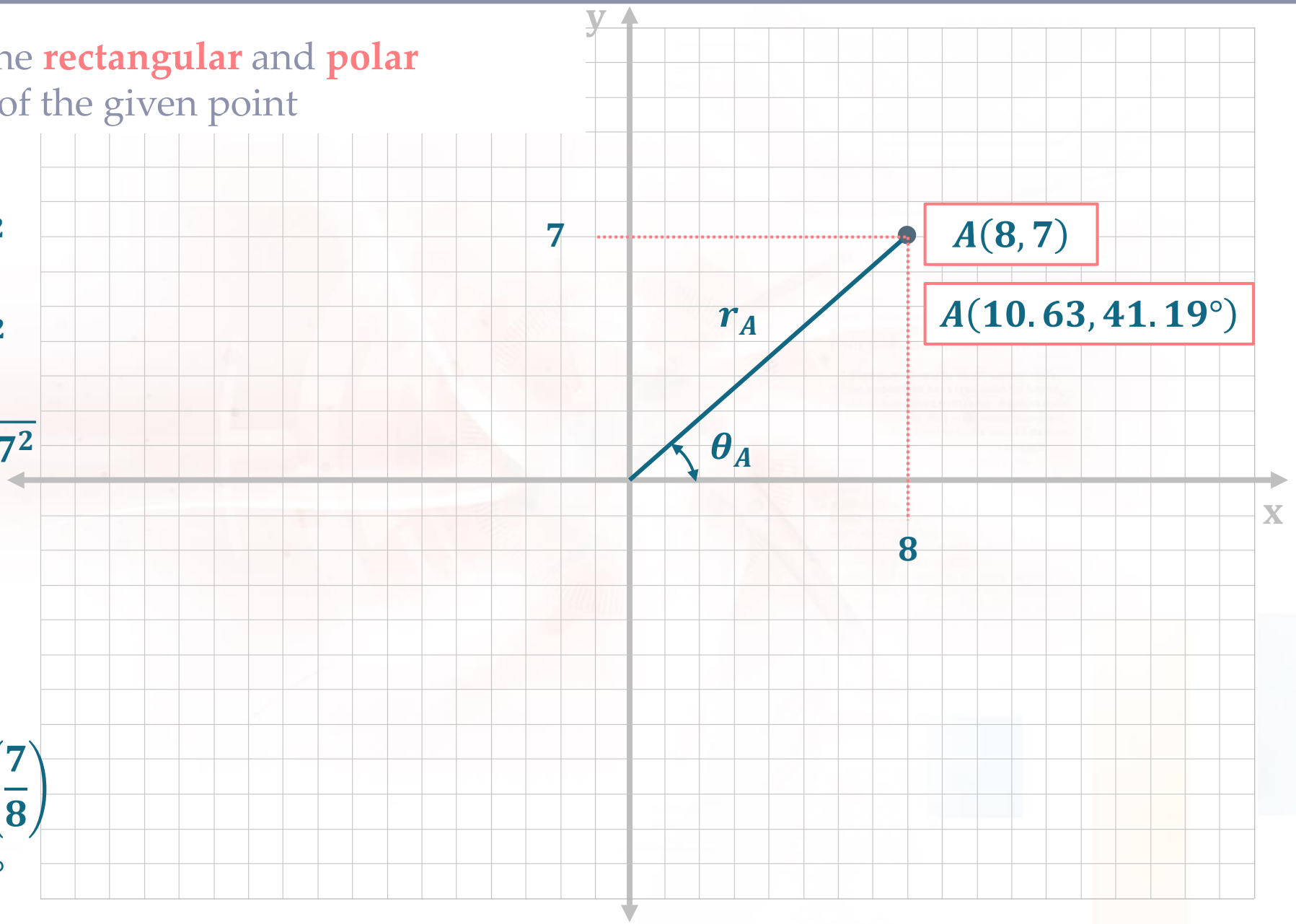
$$r_A = \sqrt{8^2 + 7^2}$$

$$r_A = 10.63$$

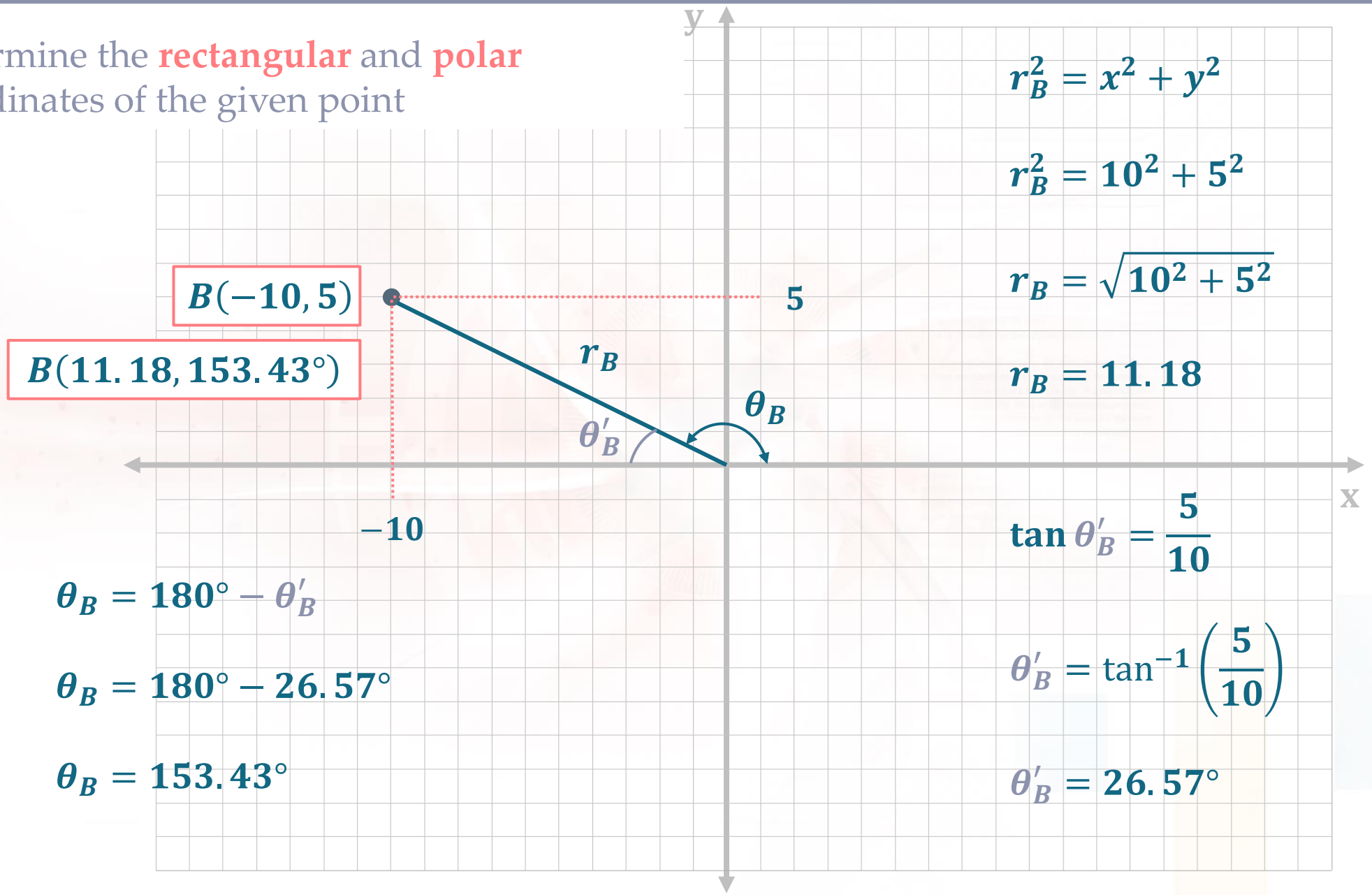
$$\tan\theta_A = \frac{7}{8}$$

$$\theta_A = \tan^{-1}\left(\frac{7}{8}\right)$$

$$\theta_A = 41.19°$$



Determine the **rectangular** and **polar** coordinates of the given point




$$r_B^2 = x^2 + y^2$$

$$r_B^2 = 10^2 + 5^2$$

$$r_B = \sqrt{10^2 + 5^2}$$

$$r_B = 11.18$$

$$\tan \theta'_B = \frac{5}{10}$$

$$\theta'_B = \tan^{-1}\left(\frac{5}{10}\right)$$

$$\theta'_B = 26.57°$$

$$\theta_B = 180° - \theta'_B$$

$$\theta_B = 180° - 26.57°$$

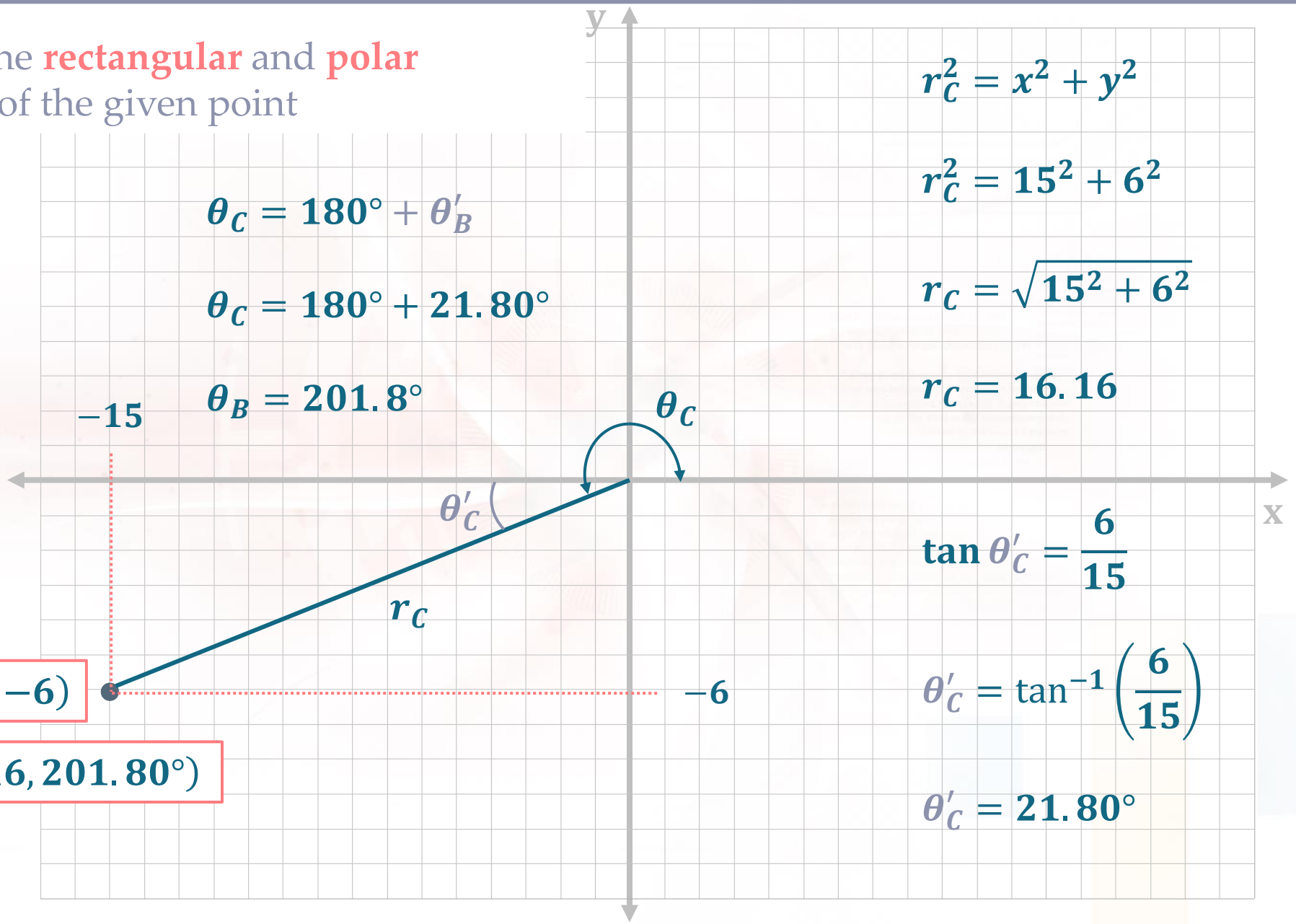
$$\theta_B = 153.43°$$

Determine the **rectangular** and **polar** coordinates of the given point

$$\theta_C = 180° + \theta'_B$$

$$\theta_C = 180° + 21.80°$$

$$\theta_B = 201.8°$$




$$r_C^2 = x^2 + y^2$$

$$r_C^2 = 15^2 + 6^2$$

$$r_C = \sqrt{15^2 + 6^2}$$

$$r_C = 16.16$$

$$\tan\theta'_C = \frac{6}{15}$$

$$\theta'_C = \tan^{-1}\left(\frac{6}{15}\right)$$

$$\theta'_C = 21.80°$$

$C(-15, -6)$

$C(16.16, 201.80°)$

Determine the **rectangular** and **polar** coordinates of the given point

$$r_D^2 = x^2 + y^2$$

$$r_D^2 = 15^2 + 10^2$$

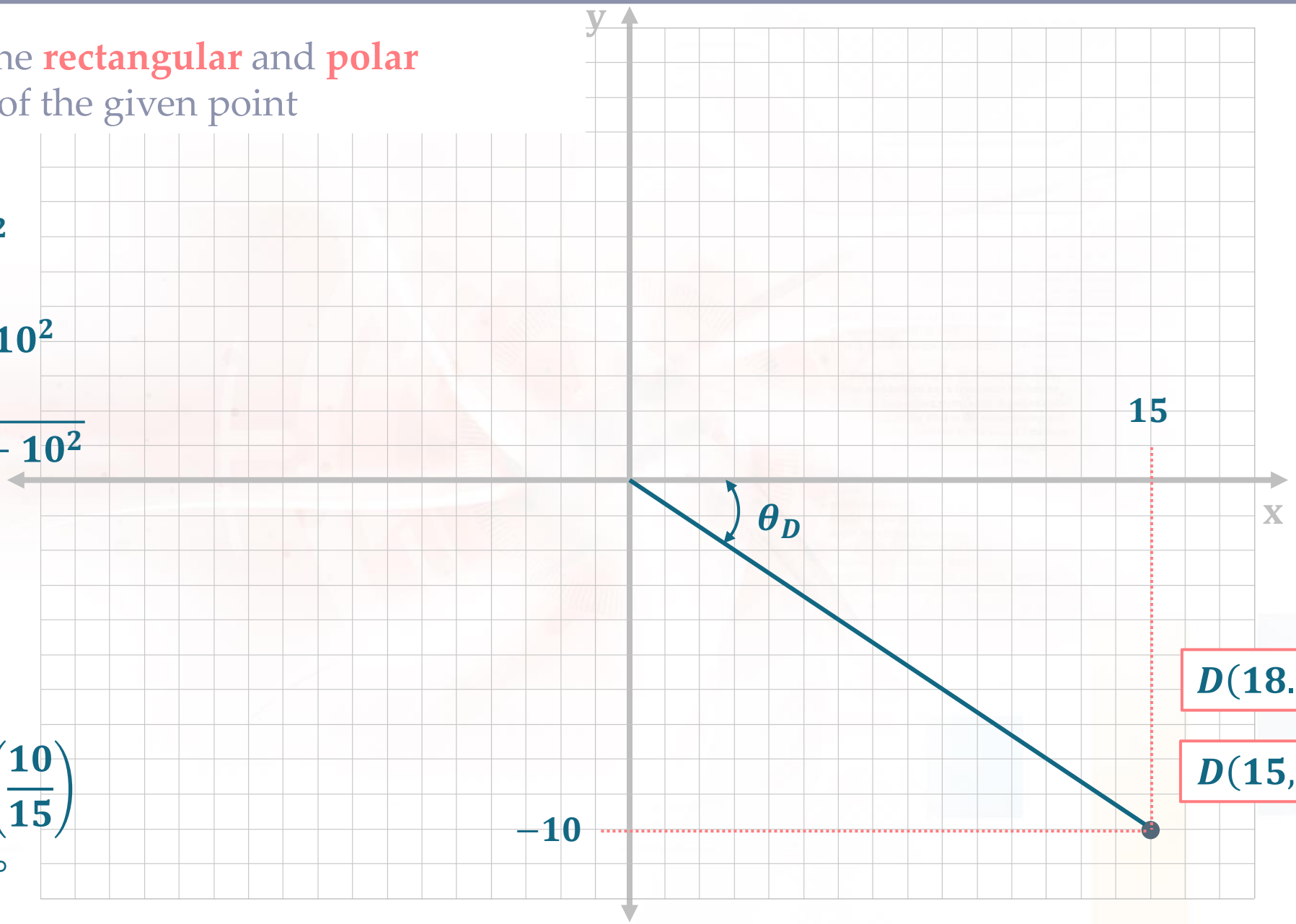
$$r_D = \sqrt{15^2 + 10^2}$$

$$r_D = 18.03$$

$$\tan\theta_D = \frac{10}{15}$$

$$\theta_D = \tan^{-1}\left(\frac{10}{15}\right)$$

$$\theta_D = 33.69°$$




$D(18.03, -33.69°)$

$D(15, -10)$

# POLAR ↔ RECTANGULAR

Given the point $\boldsymbol{P(r, \theta)}$ ,

the **rectangular** coordinates are:

$$\boldsymbol{x = r\ \cos\theta}$$

$$\boldsymbol{y = r \sin\theta}$$

Given the point $\boldsymbol{P(x, y)}$ ,

the **polar coordinates** are:

$$\boldsymbol{r^2 = x^2 + y^2}$$

$$\boldsymbol{\tan\theta = \frac{y}{x}}$$

Convert the polar coordinates $P\left(7, \frac{2\pi}{3}\right)$ and

$Q\left(19, \frac{5\pi}{4}\right)$ to **rectangular** coordinates

$$P_x = 7\cos\left(\frac{2\pi}{3}\right)$$

$$P_x = -3.5$$

$$P_y = 7\sin\left(\frac{2\pi}{3}\right)$$

$$P_y = 6.06$$

$$\boxed{P(-3.5, 6.06)}$$

*ans*

$$Q_x = 19\cos\left(\frac{5\pi}{4}\right)$$

$$Q_x = -13.43$$

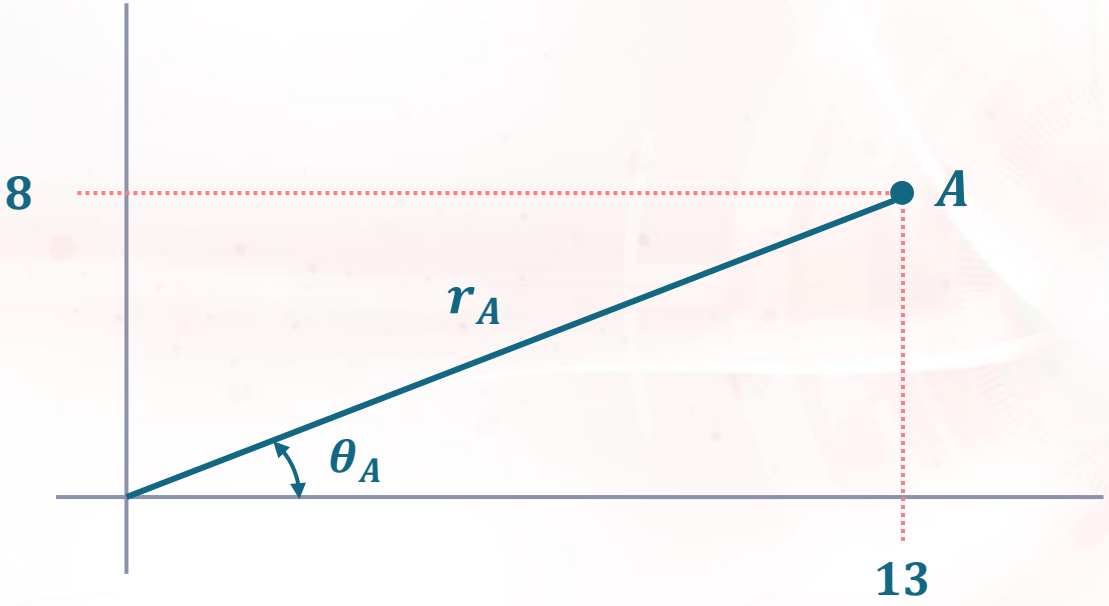
$$Q_y = 19\sin\left(\frac{5\pi}{4}\right)$$

$$Q_y = -13.43$$

$$\boxed{Q(-13.43, -13.43)}$$

*ans*

Convert the rectangular coordinates $A(13, 8)$ and $B(-17, 22)$ to **polar** coordinates




$$r_A^2 = x^2 + y^2$$

$$r_A^2 = 13^2 + 8^2$$

$$r_A = \sqrt{13^2 + 8^2}$$

$$r_A = 15.26$$

$$\tan\theta_A = \frac{8}{13}$$

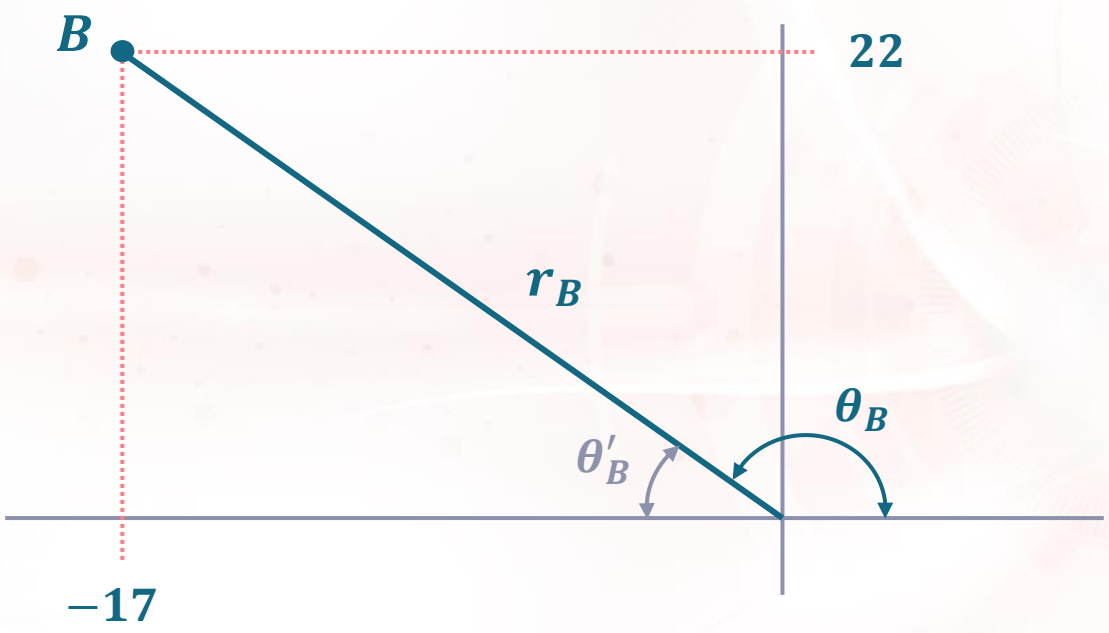
$$\theta_A = \tan^{-1}\left(\frac{8}{13}\right)$$

$$\theta_A = 31.61°$$

$$\boxed{A(15.26, 31.61°)}$$

*ans*

Convert the rectangular coordinates $A(13, 8)$ and $B(-17, 22)$ to **polar** coordinates




$$r_B^2 = x^2 + y^2$$

$$r_B^2 = 17^2 + 22^2$$

$$r_B = \sqrt{17^2 + 22^2}$$

$$r_B = 27.80$$

$$\tan\theta'_B = \frac{22}{17}$$

$$\theta'_B = \tan^{-1}\left(\frac{22}{17}\right)$$

$$\theta'_B = 52.31°$$

$$\theta_B = 180° - \theta'_B$$

$$\theta_B = 180° - 52.31°$$

$$\theta_B = 127.69°$$

$$\boxed{B(27.80, 127.69°)}$$

*ans*

# SEATWORK